

Shifting Perspectives on Renormalization

The remarkable quantitative agreement of the QED amplitudes ... computed using this procedure with the measured experimental values remain among the most impressive successes of physical science. Nevertheless, the overpowering impression persisted among many physicists that the procedures of Dyson amounted to an intellectually unsavory “fudge” — a mere sweeping under the rug of potential underlying inconsistencies in the theory of which the ultraviolet divergences were apparently the overt symptom. (Duncan 2012, 611)

- What is the justification for renormalization techniques, granting that they are essential to success of QFTs? (Note that in particular we lack non-perturbative treatment of these issues that would help to clarify status of perturbative treatments.)
- Is renormalizability a necessary condition for a physically reasonable theory?

Making the Connection with Observations

Very schematically, treating the interaction H_I as a perturbation to free field theory governed by H_0 (“interaction picture”) we calculate quantities such as S -matrix elements via an expansion:

$$\langle out|\hat{S}|in\rangle = \langle out|\hat{1}|in\rangle + i\langle out|e^{-iH_I t}|in\rangle \quad (1)$$

$$(2)$$

Calculate this by expanding $e^{-iH_I t}$ term by term, in increasing powers of H_I (and the coupling constant). In the case of QED, for example, we end up with an expansion in powers of the coupling constant α :

$$A_0 + \alpha A_1 + \alpha^2 A_2 + \dots \quad (3)$$

Feynman diagrams are used to generate and calculate the terms in this expansion, the overall contribution A_i at a given order includes several distinct terms; the expansion includes “all connected, amputated diagrams” constructed according to rules based on the free field and interaction. (Kinoshita 2014)

One of the first successes of QED was Schwinger’s calculation, based on the finding the first term in an expansion of this form (the one-loop correction to the electron vertex function) for the anomalous magnetic moment of the electron (see Koberinski and Smeenk 2020, for more). At each order one has to include all the relevant processes (Feynman diagrams) to find the overall A_i , with the number of diagrams increasingly rapidly as n increases. (For example: Kinoshita calculates up to $n = 10$, which involves 12,672 diagrams!)

Infinites

Problems for this kind of perturbative expansion, to be addressed by renormalization:

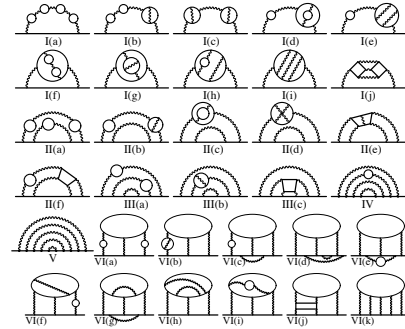


FIG. 2. Typical self-energy-like diagrams representing 32 gauge-invariant subsets contributing to the tenth-order lepton $g-2$. Solid lines represent lepton lines propagating in a weak magnetic field.

- Divergences for individual terms, due to loop integrals over unrestricted momentum. *How can we find a finite value for each of integrals included in A_i ?*
- Divergence of the series expansion itself: Dyson's (1952) argument; number of diagrams increases too fast with n . *How should we understand the perturbative expansion; should we take this as merely an approximation (Fraser 2016), or does this require some novel semantics (see Miller 2021)?*
- Consistency of the overall framework (Haag's theorem and the use of the "interaction picture", treating full theory as perturbation to free field theory)
- ∞ and vacuum energy: from the analogy with simple harmonic oscillators, expect that the ground state for QFT has a non-zero energy for each mode k : $\frac{1}{2}\hbar\omega_k$ (sometimes called the zero-point energy). Integrating this leads to infinite energy. Usual response: simply ignore this in QFT, since predictions only depend on relative energy differences.
The contrast: the issue arises due to the *order of the operators* ($\hat{a}^\dagger, \hat{a}$) in the expression for the Hamiltonian, but this is not fixed by the procedure of quantization. We can "normal order" the operators to eliminate the zero-point energy. (See Teller, pp. 128-130.)

Regularizing and Renormalizing

Regularization: mathematically alter the divergent integrals, usually by introducing some new parameter(s) such that: the integral is finite for finite values of the parameter, and the original integral is recovered in an appropriate limit.

- Conceptually simplest form of regularization: introduce an "ultraviolet cutoff" Λ , in effect imposing an upper limit on the momentum. The terms that lead to divergences have the form $\int_0^\infty \frac{d^4k}{k^4}$ (logarithmic divergence) are then replaced with $\int_0^\Lambda \frac{d^4k}{k^4}$ (finite for finite Λ).
- Note that there are a wide variety of different regularization techniques. For example, dimensional regularization takes the number of spacetime dimensions to be d , such that we are considering the integral $\int_0^\infty \frac{d^d k}{k^4}$ and recover the original in the limit as $d \rightarrow 4$.

(One general issue: we cannot make this change without altering the features of the theory, such as symmetries; there is a pragmatic question regarding how to balance calculational ease vs. the impact of these changes.)

Renormalization: rewrite the original Lagrangian in terms of "physical" or "renormalized" parameters, and impose "renormalization conditions" at a specific scale. For a renormalizable theory this will eliminate *all* the divergences, leaving finite expressions in the perturbative expansion.

- The original Lagrangian includes a number of "bare parameters." For example, given the Lagrangian for a simple scalar field theory (the last term is the interaction):

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \quad (4)$$

there are three "bare parameters": ϕ, m, λ .

- We rewrite the Lagrangian in terms of new parameters (this is merely algebra!), introducing “renormalized parameters” ϕ_R, m_R, λ_R . This leads to a new Lagrangian (for the standard way of re-scaling):

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_R)^2 - \frac{1}{2}m_R^2 \phi_R^2 - \frac{\lambda_R}{4!} \phi_R^4 \quad (5)$$

$$+ \frac{1}{2}\delta_z(\partial_\mu \phi_R)^2 - \frac{1}{2}\delta_m \phi_R^2 - \frac{\delta_\lambda}{4!} \phi_R^4 \quad (6)$$

where the expressions in the second row include the “counterterms.” The counterterms are “shifts” away from the bare values, e.g. $\delta_m = m_B^2 Z - m_R^2$. (As Fraser emphasizes, merely algebraic shifts in the parameter often do change the properties of a power series expansion.)

- *Renormalization conditions:* we further use empirical input to make these definitions more useful. In particular, we assume that the theory applies to specific kinds of scattering experiments; granting this assumption, we can directly measure the values of the renormalized parameters at some specific energy scale.
- The key point is that the rescalings, given the regularization scheme and renormalization conditions, allow us to cancel out the divergent parts of the original integrals and then use the QFT formalism to calculate a number of other quantities. For cut-off regularization, for example, we start with Λ finite and define the counterterms. As we take $\Lambda \rightarrow \infty$, we have two infinite quantities that cancel out and leave finite expressions that are functions of the “renormalized parameters” ϕ_R, m_R, λ_R (at some specified energy scale). (This is just a more general version of what Teller describes for m on p. 155-157.)
- *Renormalizability:* There is a natural question that arises, given that we are approaching this iteratively: suppose we find a set of counter-terms that can be used to eliminate all divergences up to n -th order in the expansion. Are there *new kinds of divergences* that arise at higher order? Do we need a *finite* number of counter-terms, or do we need to introduce new counter-terms at each order?

Perturbatively renormalizable theories: the initial set of counter-terms suffice to eliminate divergences at *all orders* in the perturbative expansion.

References

- Kinoshita, T. (2014). Tenth-order qed contribution to the electron $g-2$ and high precision test of quantum electrodynamics. In *Proceedings of the conference in Honour of the 90th Birthday of Freeman Dyson*, pages 148–172. World Scientific.
- Koberinski, A. and Smeenk, C. (2020). Q.E.D., QED. *Studies in the History and Philosophy of Modern Physics*, 71:1–13.